

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer — Remote

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Professional Summary

Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data science solutions to solve complex challenges. Expert in managing complete project lifecycles, focusing on stakeholder engagement, KPI definition, development, team building, and deployment.

Education

Bachelor of Science in Mechanical Engineering

2006

Arizona State University — Tempe, AZ

STEM Coursework: Computational Thermodynamics, Design of Experiments (DoE), Differential Equations, Economics, Finance, Finite Element Analysis, Linear Algebra, Logic, Manufacturing, Material Science, Multivariate Calculus, Numerical Methods, Statistics

Skills

Hard Skills: AI/ML, Agile, Anomaly Detection, Architectural Design, Artificial Intelligence, AWS, CI/CD, Continuous Integration, Database Administrator, Data Gathering, DBA, Deep Learning, DevOps, Developer Tools, Distributed Computing, Docker, ECR, Embedding, ETL, Event Bridge, Git, GitHub Actions, GitHub Copilot, Hugging Face LLMs, Large Language Models, Power BI, Problem Solving, Python (Autogluon, Pandas, PyTorch, Scikit-learn, TensorFlow), R, Redshift, Research, S3, Sagemaker, Snowflake, SQL, Step Functions, Streamlit, Statistical Tools, Tableau, Transformers

Soft Skills: Adaptability, Business Requirements, Coaching, Creative, Growth Mindset, Leadership, Mentoring, Strategic Thinking, Willing to Learn

Experience

Lead Data Scientist

Sept 2022 – Present

Caterpillar Inc. — Surprise, AZ

- Architect and developer of end-to-end development of high-impact data science initiatives, including an award-winning AI system to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award.
- Partnered with leadership and stakeholders to map the business problems the AI solves, as well as the legal constraints to consider. Worked to maintain alignment within the greater enterprise.
- Wrangled structured, semi-structured and unstructured data to compile a training dataset from multiple sources. Thousands of parameters were considered derived from dealer and product information. An LLM was used to find similarities in job descriptions. Feature selection was used to keep important parameters.
- Utilized Python and Autogluon to train AI model by leveraging machine learning frameworks like neural networks, PyTorch, XGBoost, TensorFlow and LightGBM.
- Engineered and deployed robust MLOps pipelines using AWS and Snowflake, as a database administrator, and established best practices for cloud and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.

Reliability Analytics Manager

Amazon — Tempe, AZ

Feb 2022 – Sept 2022

- Pioneered design engineering KPI for autonomous robots by analyzing petabyte-scale IoT data. Authored an influential white paper, as reproducible research using R, on the framework for performing Weibull analysis and secured adoption of the new method by senior leadership.
- Designed and optimized winning regression and recommendation systems, achieving two-time champion status in Amazon Machine Learning University competitions.

Data Engineering Manager

Empire-Cat — Mesa, AZ

Jan 2012 – Feb 2022

- Created ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days.
- Developed and deployed an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting.
- Directed a high-performance team of up to 8 engineers and functioned as client facing subject-matter-expert for mining equipment. Leadership success by fostering a data-driven approach to solve equipment performance issues as Reliability Engineering Manager before being promoted.
- Resolved acute and chronic design flaws, operation, remanufacturing and maintenance opportunities for large off-highway truck fleet through DMAIC 6 Sigma with analysis in R and SQL, saving clients \$1.2 billion over six years.

Certifications

Arizona State Board of Technical Registration: EIT, Mechanical Engineering, Registration No. 10670

BerkeleyX: The Science of Happiness at Work Professional Certificate

Canonical: Ubuntu Linux Professional Certificate

CSCMP: Supply Chain Foundations Demand Planning Professional Certificate

Data Camp: AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate

Empire-Cat: Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations

GitHub: Career Essentials in GitHub Professional Certificate

IUOE Local 150: Diesel Mechanic, Bulldozers, Class B CDL, Cranes, Forklifts, Heavy Equipment Operator, MSHA, OSHA, Preventive Maintenance, Scrapers, Skid-steers, Telehandlers, Wheel Loaders Shielded Metal Arc Welding (SMAW)

Komatsu North America: Certified Technical Communicator

Society of Automotive Engineers: Cost and Finance for Engineers

Udemy: Streamlit for Snowflake Masterclass